Charge Determination for the Tritium Experiments

Hall A, Jefferson Lab

Analysis report

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Abstract

The beam current monitor system is used to measure and control the amount of charged particle beam current delivered to the target. The system is composed by two cavity monitors (BCMs) and a parametric current transformer (PCT) known as the Unser. This report presents an overview of the current monitor system, along with the calibration process required and followed during the tritium set of experiments in the 2017-2018 years.

Contents

1 Introduction **1**

2 Current Monitor System **1**

2.1 Unser monitor 2

2.2 RF cavity monitors 3

3 Linearity Check with an RF Signal Generator **5**

4 Unser Wire Calibration **9**

5 BCM calibration **13**

6 Charge calculation **18**

Look at Only these

List of Figures

1 Beam current monitor instruments in Hall A. Taken from [1]. 1

2 Unser signal flow to the DAQ system at Hall A. 2

3 Resonant RF cavity. 3

4 BCM signal flow. The upstream cavity only has 1 gain amplification. 4

5 u1 linearity test result. 6

6 d1 linearity test result. 6

7 d3 linearity test result. 7

8 d10 linearity test result. 7

9 unew linearity test result. 8

10 dnew linearity test result. 8

11 Unser background: simulation of the Unser background with $I = 0$. . 11

12 Unser calibration process. a. Local offset subtraction, b. Unser rate peak averages, and c. the source current with respect to the Unser rate average points , with a Linear fit. 12

13 Unser calibration, Runs 662 and 90436. 14

14 BCMs rates, Fall 2017. 15

15 dnew BCM calibration. 16

16 Current profile in an average run. 18

List of Tables

1 BCMs current ranges with good linearity for the analog and digital receivers. 9

2 BCM calibration constants from December 2017 to September 2018. . 17

1 Introduction

The purpose of this document is to present the method that was used to calculate the charge in the tritium set of experiments performed during the 2017-2018 years. First, the description of the charge monitor is done, and finally, the calibration process and the method to calculate the charge are shown.

2 Current Monitor System

The beam current monitor system consists of a toroidal sensor (Unser), an RF cavity in the upstream side and an RF cavity in the downstream side of the Unser, and a data-acquisition system. The Unser has a sensitivity of 4 $mV/\mu A$ [1], but the offset drifts over time. The large offset noise vetoes the possibility of employ the Unser during all the measurements. Therefore, the Unser is an absolute measurement of the current used to calibrate the BCMs. These calibrations are required periodically.

PID used to set the temp? Have seen this before?

The beam current monitor instruments are enclosed in a thermal insulation box kept at $110F$, the thermal insulation reduces the noise and drift in the Unser threshold and avoid miss-tuning in the resonant cavities. In addition, the box has three magnetic shields, two of iron and one of μ material, as shown in Figure 1. In the hall, the box is located 25 m upstream of the target [2].

Why is cavity so tall?

The cavity holds an antenna

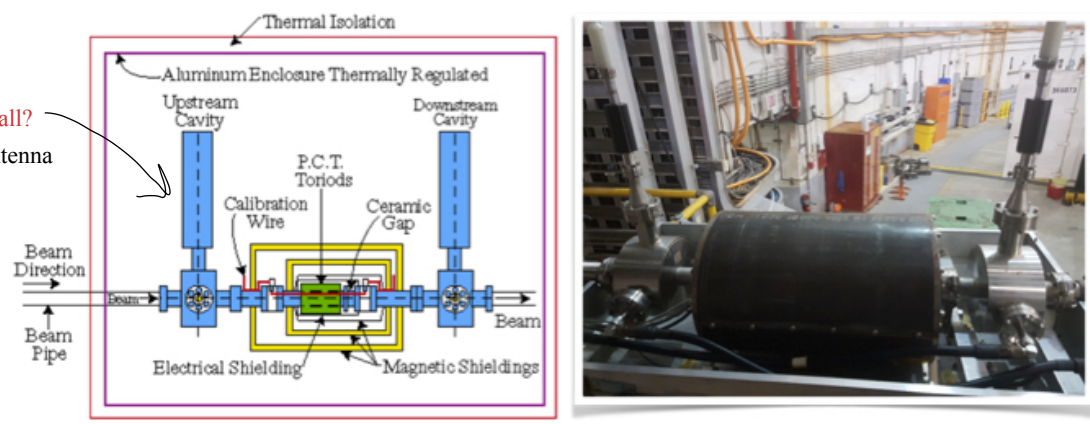


Figure 1: Beam current monitor instruments in Hall A. Taken from [1].

In the center of the box is the Unser with two ceramic gaps in each side, the gaps

prevent parasitic currents from the beam pipes. In each side of the Unser, an RF cavity is located (upstream and downstream), tuned to 1.497 GHz [1].

2.1 Unser monitor

The Unser monitor was named after its inventor K. Unser in 1969 at Cern [3]. It is composed of two identical toroidal cores driven in opposite ways by an external source. The DC component of the current flowing through the toroid sensor is detected by a magnetic modulator. The beam current in the cores produces a flux imbalance, which generates an output signal proportional to the even harmonics of the frequency of excitation, In the absence of DC current, the sum of the signals is zero [1].

The Unser is calibrated by sending a known DC current through the wire that passes inside of the toroids (See Figure 1) with a current source. The Keithley Precision DC low noise current source, model 6220, is located in the Counting House and is used for the calibration.

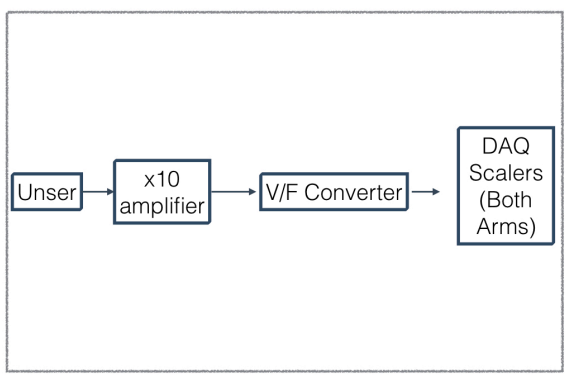


Figure 2: Unser signal flow to the DAQ system at Hall A.

Since the Unser is very sensitive to magnetic and temperature fluctuations, the offset drifts over time, while the gain remains stable within 0.1 % of 4 $mV/\mu A$. Additionally, the Unser signal is sent to the Counting House, where it goes to an amplifier and a Voltage-to-Frequency (V/F) converter, to be finally sent to both DAQ system in LHRS and RHRS huts, as shown in Figure 4.

This also creates more noise

Therefore, the Unser calibration is required to be performed regularly during the experiments, to monitor the full performance of all the equipment and the information chain. This information is used as the absolute calibration of the resonant RF cavity monitors, known as BCMs.

2.2 RF cavity monitors

Two stainless steel resonant cavities surround the Unser and are tuned to the frequency of the beam 1.497 GHz with a quality coefficient $Q \approx 500$. The cavities are composed of loop antennas located where the magnetic field is maximum. When the beam passes through the output RF signal is proportional to the current [1].

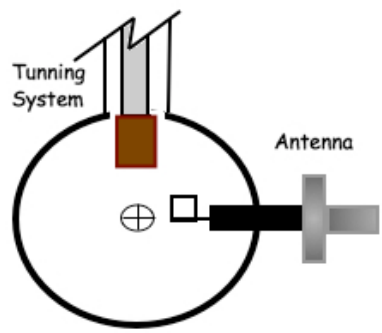


Figure 3: Resonant RF cavity.

The BCMs must be kept at a constant temperature within 0.2 $°C$, which induces an error of 0.07%. and the effect of the beam displacement about 0.06% error.

As shown in the Figure 4, the output signal from the BCMs has a frequency of 1497 MHz, which is sent through 7/8" heliax cable (to avoid attenuation), to a splitter, that goes into two paths: i) goes to a 1 MHz downconverter, which sends the signal to a set of amplifiers with gains 1, 3 and 10 to cover the whole dynamic range for each cavity. Next, the signal is sent to a V/F converters and then to the scalers in the DAQ systems of both (LHRS and RHRS) arms. Currently, the upstream cavity has only a gain 1 amplifier, while the downstream cavity keeps the three gains. A copy of the 1 MHz attenuated signal is also sent to the HP3458 digital voltmeters

for Epics. And ii) the digital receivers designed by Musson group at Jefferson lab [4], which can cover the whole range with high linearity.

The digital receivers are used with the analog output, the signals go to the V/F converters that were used with the gains 3 and 10 of the upstream cavity, which are finally sent to the scalers in both arms. On November of 2017, the upstream signal in the digital receiver had an attenuation of 25 *db* and the downstream of 20 *db*.

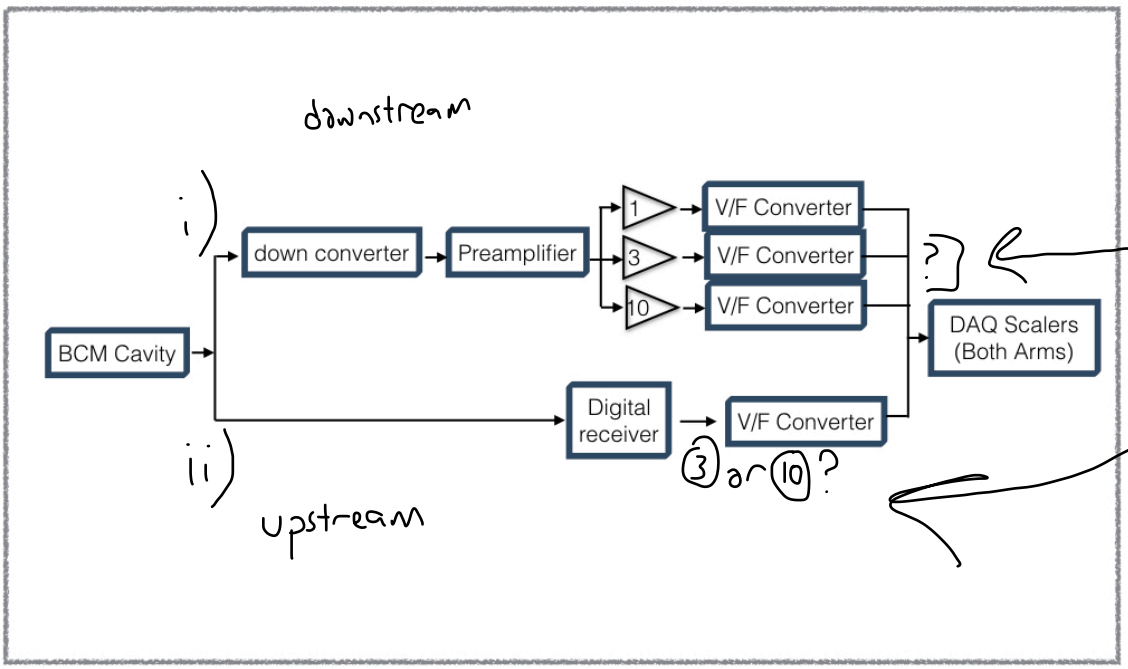


Figure 4: BCM signal flow. The upstream cavity only has 1 gain amplification.

Furthermore, in the DAQ system, the upstream signals are labeled u1 for $1x$ amplification and unew for the digital receiver, and the downstream signals are labeled d1, d3 and d10, for the $1x$, $3x$ and $10x$ amplifications, respectively, and dnew is the digital receiver signal (See Figure 4).

3 Linearity Check with an RF Signal Generator

The study of the current range for each BCM was performed using a function generator during the Fall of 2017.

The function generator was used to simulate the nominal cavity output power, where 40 dBm correspond to 1 μA. It was connected to the splitter at the cavity output, which sends the signal to the analog system and the digital receiver.

It is necessary to consider the attenuation due to the heliax cable between the signal generator and the splitter, which corresponds to -0.065 dBm. Therefore, to get the equivalent of 1 μA of output power at the splitter, the signal generator was set a little higher (at -39.35 dBm).

Using the definition $P(dBm) = 10log(I^2 Reff/0.001Watt)$, and the above power level at 1 muA, the current at any other power level is given in μA by,

$$I = 10^{(P/20)} \cdot 92.7897 \tag{1}$$

Finally in order to get a full set of data points, steps of 5 dBm were taken from 0.65 dBm to -64.35 dBm, getting data points for the currents: 99.99, 56.23, 31.62, 17.78, 10.00, 5.62, 3.16, 1.77, 1.00, 0.56, 0.31, 0.17, 0.1 and 0.05 μA.

The corresponding run numbers for this study are 90206, 90207, 412 and 413, for the Right and Left HRS, and the analysis graphs are in Figures 5, 6, 7, 8, 9 and 10.

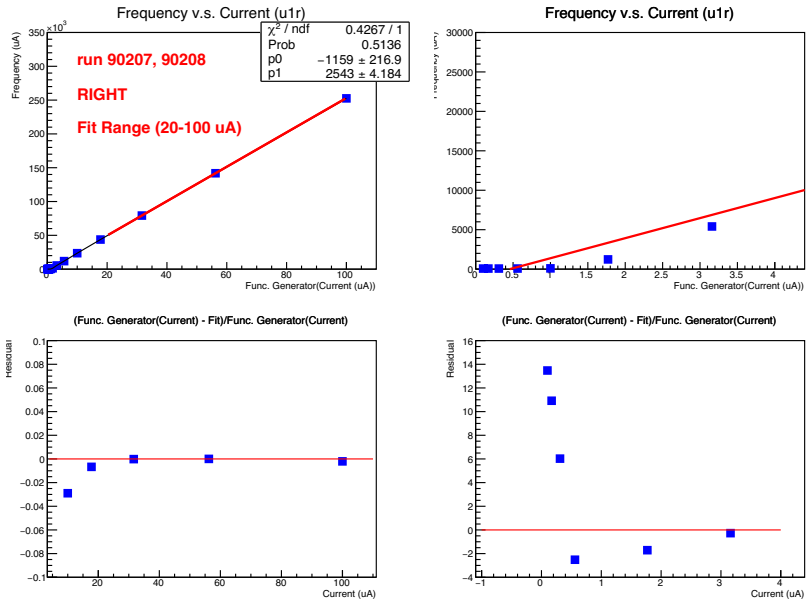


Figure 5: u1 linearity test result.

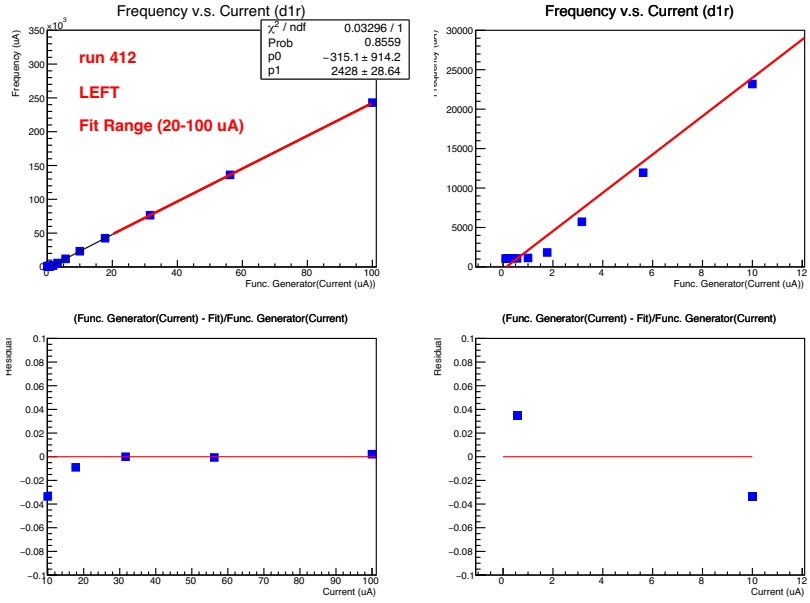


Figure 6: d1 linearity test result.

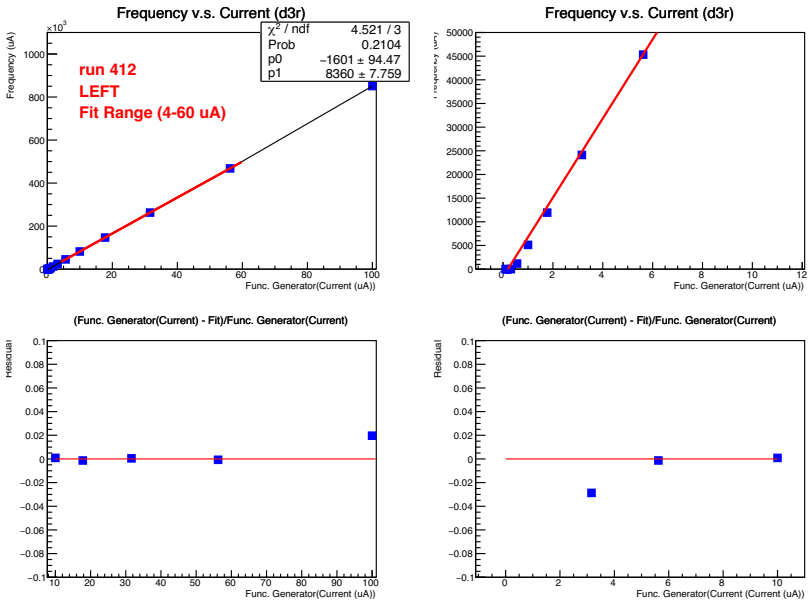


Figure 7: d3 linearity test result.

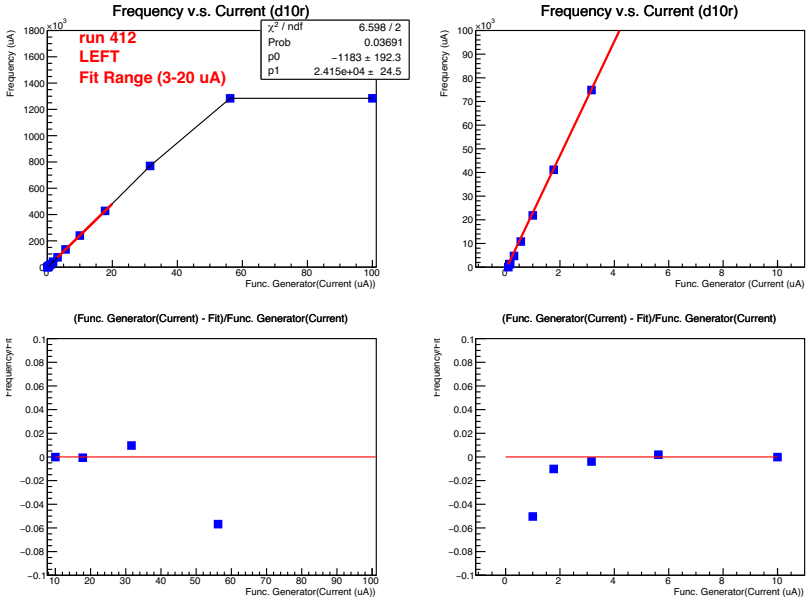


Figure 8: d10 linearity test result.

As result of the linearity test, the summary of current ranges for each BCM signal

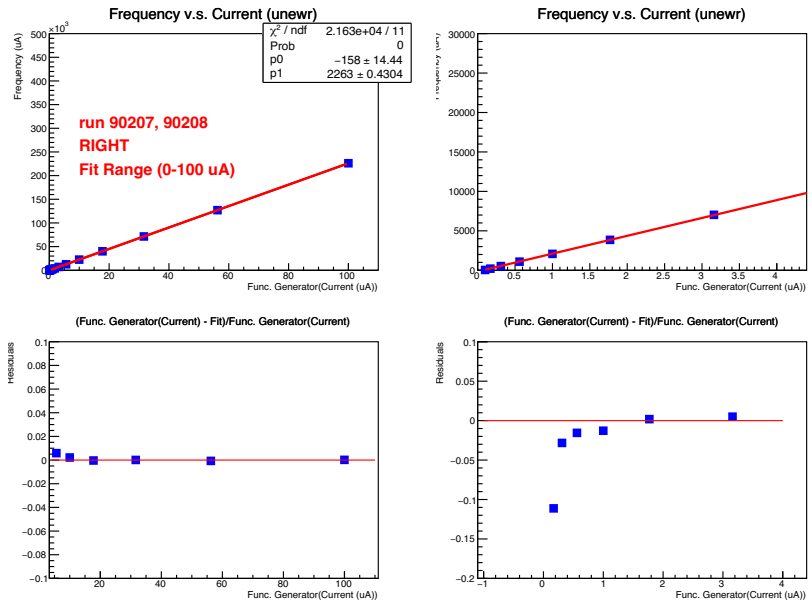


Figure 9: unew linearity test result.

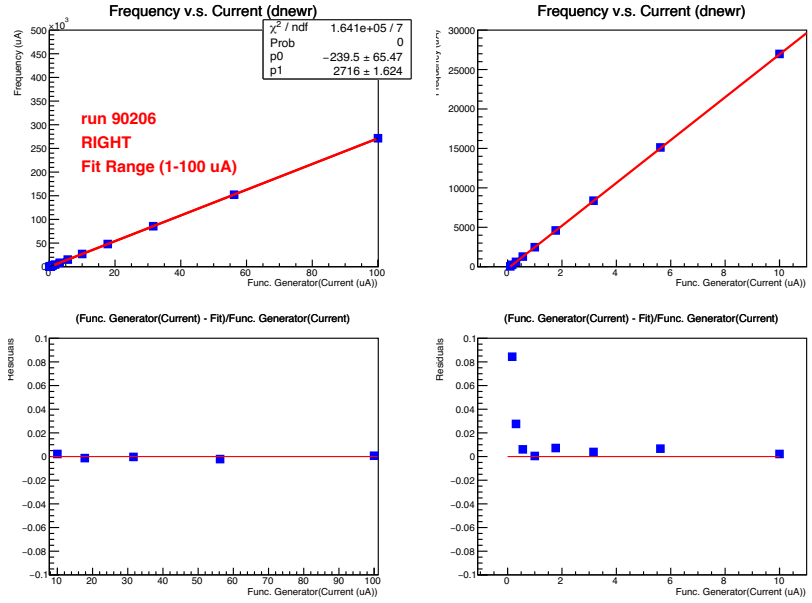


Figure 10: dnew linearity test result.

is in the Table 1. Therefore, since the Tritium target has a current limit of 22.5 μA, the BCMs available to calculate the charge are d3, d10, unew and dnew, and the BCMs recomended for the luminosity scans are unew and dnew.

BCM	**Current Range** (μA)
u1	30-100
d1	30-100
d3	5-60
d10	3-25
dnew	1-100
unew	2-100

Table 1: BCMs current ranges with good linearity for the analog and digital receivers.

4 Unser Wire Calibration

The Unser calibration is done using the Keithley Precision DC low noise current source, model 6220, located in the Counting House. There are three different alternatives to communicate with the current source:

1. Manual calibration: using the front panel of the current source. The user has the responsibility to measure the time and to change the current values in the planed scale for the calibration.

2. The Epics interface: found the Hall A menu (Current monitor calibration source). In normal operation the source is in Standby, to start the calibration put 0 in the Current label and press operate. After the source in operation mode, just put the desired current and hit enter. At the end of the calibration, the current source must come back to standby mode.

3. If the user has access to an accelerator account, a simple script in any language can be used to calibrate the Unser. However, when the accelerator is in beam mode, the access is blocked. Then, permission should be asked in order to use these commands lines. An example of a python code is:

```
from time import sleep
from subprocess import call
```

```
from subprocess import check_output
import matplotlib.pyplot as plt
import numpy as np
loop=8
highcurrent=100
lowcurrent=0
readback=[]
#
# Set Low Current And Turn On
#
call(["caput","HA:K224:Current","
call(["caput","HA:K224:Operate","1"])
#
# Set Currents Between High and Low
#
for l in range(loop):
if l
call(["caput","HA:K224:Current","
if l
call(["caput","HA:K224:Current","
sleep(1)
a = check_output(["caget","HA:K224:Current"])
print(a)
sleep(1)
#
# Turn Everything Back Off
#
call(["caput","HA:K224:Current","0"])
call(["caput","HA:K224:Operate","0"])
```

Preparing for the calibration requires that CODA can be run for both arms, the current source is up and working, and the Unser is responding to the change in currents. This last requirement can be done by checking response in the rate of the Unser and/or checking the response in the Unser multimeter for the different currents.

First the LHRS and RHRS coda runs must be started, to record the scalers, use the clock to take reasonably high data rate (above about 50 Hz at least) We used the

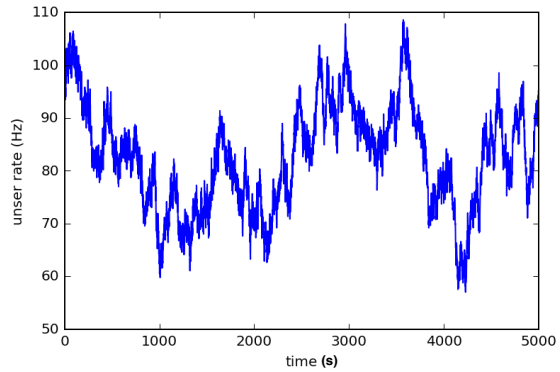


Figure 11: Unser background: simulation of the Unser background with $I = 0$.

Left clock with a prescale factor of 500. (PS8 = 500). The analysis will be done using the fast scalers.

Then, the Keithley Current source sends a known current to the Unser wire. In the first Unser calibration in Fall 2017, the set of current values were: 0, 10, 0, 25, 0, 50, 0, 75, 0, 100 and 0 (μA). Each current value is sent by 90 seconds. The 0 μA step for each current ($I > 0$) is used to subtract the local background (i.e. to subtract the pedestal) of the Unser signal. Furthermore, the set of currents is sent two times to the Unser to have better statistics.

After collecting and replying the data, the root files are used to calibrate the Unser. In the scaler trees in the root file are: TSLeft(TSRight) and evLeft(evRight), where the TSLeft(TSRight) are known as the slow scalers, they are recorded with the epics variables every 8 seconds; and the evLeft(evRight)are known as the fast scalers, which are recorded every 100 events. The fast scalers are used to calibrate the data. Specifically, the variable used to calibrate the Unser is the rate: $evLeftunser_r$.

he offset is subtracted, for each current value ($I > 0$). In other words, the pedestal average, before and after the current peak (when current = 0 μA) is subtracted from each current peak.

An average of all the 0 μA data points cannot be used in the offset subtraction, since the Unser offset is similar to the random walk and every peak of current may not have the same level of background (As shown in Figure 11).

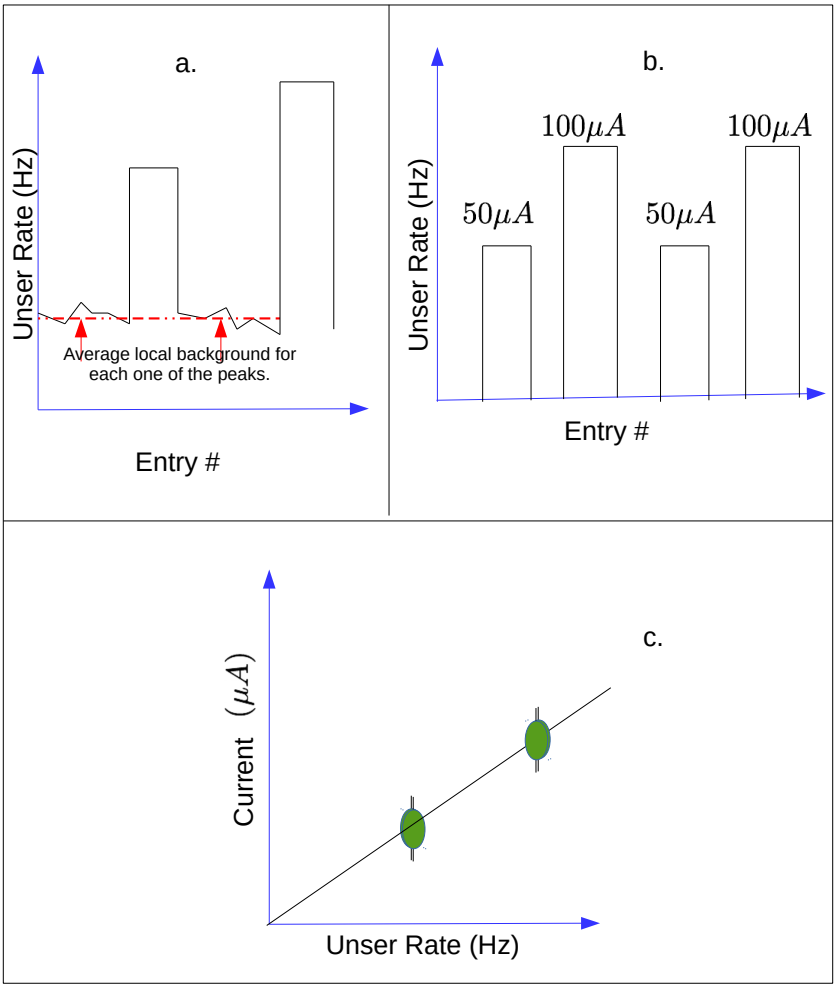


Figure 12: Unser calibration process. a. Local offset subtraction, b. Unser rate peak averages, and c. the source current with respect to the Unser rate average points , with a Linear fit.

The goal of the Unser calibration is to obtain the gain, therefore, the local offset is subtracted. Figure 12a. shows how the local offset of each peak must be averaged

and subtracted from the peak. A subtraction needs to be done for each one of the peaks.

After the pedestal subtraction (See Figure 12b) the current value of the source is plotted with respect to the average frequency of the peaks for each current, and a linear fit is done over the data points (See Figure 12c). The error in the unser rate response is given by $\frac{1.5\mu A}{\sqrt{time}}$, where $time = 90s$.

Figure 13 shows the unser calibration done with the fast scalers for the Left and Right HRS, with the run numbers 662 and 90436. The gain obtained with the LHRS daq was: $g_{LHRS} = 0.0002497 \pm 9.648 \times 10^{-7}\ \mu A/Hz$, and with the RHRS was: $g_{RHRS} = 0.0002489 \pm 9.641 \times 10^{-7}\ \mu A/Hz$.

5 BCM calibration

Before to start the calibration, it may be useful to have someone over in MCC to help to coordinate the calibration with them. And, since the calibration is done with the beamline equipment, the target is not relevant. Take data on carbon target (or any target you find useful). The calibration of the BCMs, only uses the BCMs scaler information, independently of the physics events.

The BCMs response is linear with respect to the current. In the calibration, a set of currents will be used to calculate the linear dependence. The tritium target has a maximum limit of 22 μA, therefore the d1 and u1 BCMs could not be calibrated.

An example of set of currents are: 0, 3, 0, 5, 0, 7.5, 0, 10, 0, 15, 0, 20, 0, 22.5, 0 μA. Each beam period (with or without current) should last 90 s. If there is a trip during a beam up period, take 90 s of beam off and restart that beam up period. The set of currents are repeated 2 times.

BCM Rates
↓
Calibrated with current
↓
Current is calculated from Unser signal
↓
Unser was calibrated previously pg 2
↓
Absolute value of current can be found

pg1 5

The BCMs rates are shown in Figure 14 for the calibration in Fall 2017. Each peak value for the rates is averaged and calibrated with respect to the current calculated with the Unser signal.

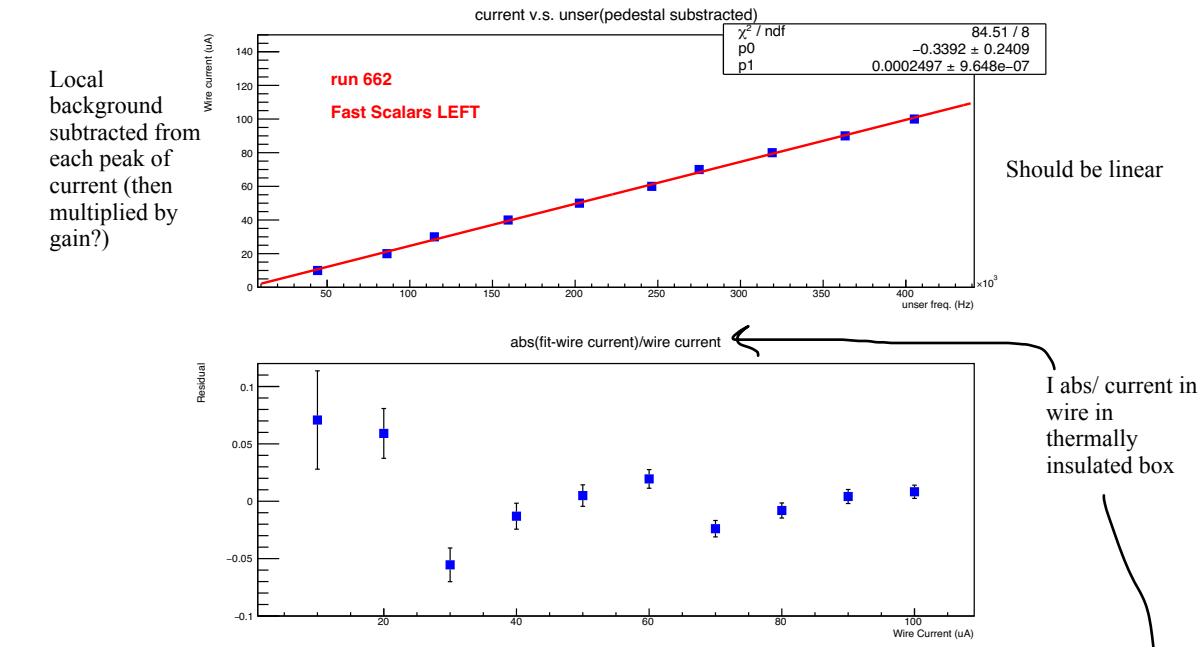
Local background = offset?

Consequently, using the calibration of the Unser obtained previously, the absolute value of current can be calculated. Therefore, the local background needs to be subtracted for each peak of current in the Unser signal, and the gain times the average of the current peak is the current value of the beam. The error of each

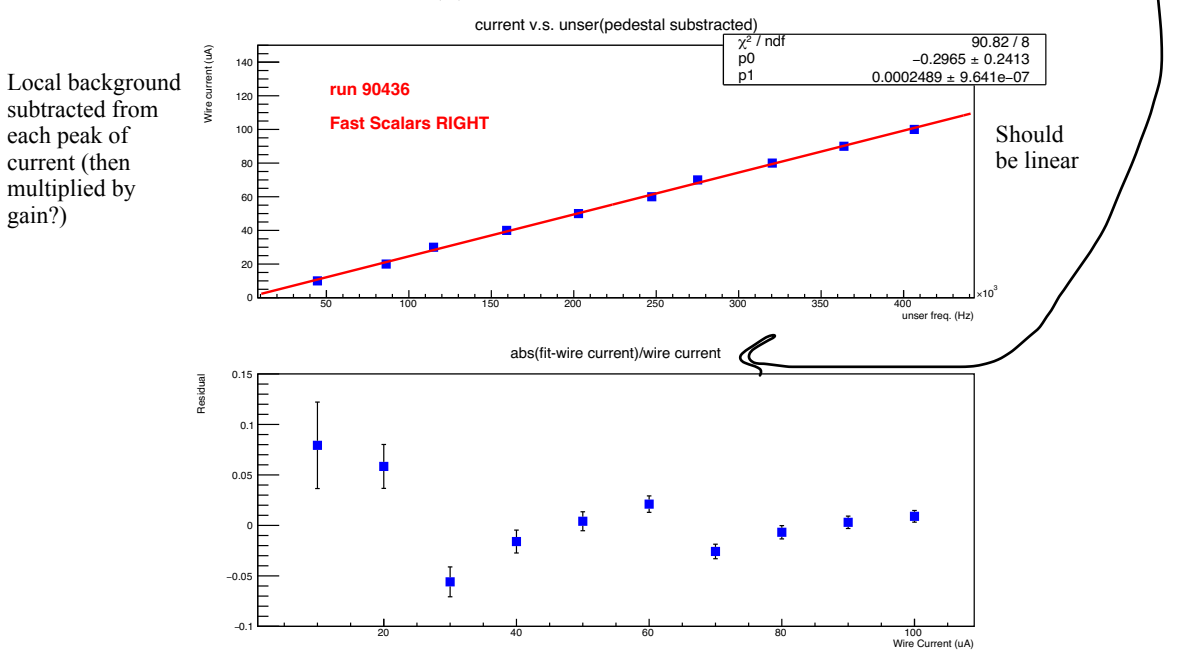
?

Or is it I absolute? Not this? I of unser = (freq measured - offset)gain of unser

I of bcm = (freq measured - offset of bcm)gain of bcm



(a) LHRS Unser Calibration.



(b) RHRS Unser Calibration.

Figure 13: Unser calibration, Runs 662 and 90436.

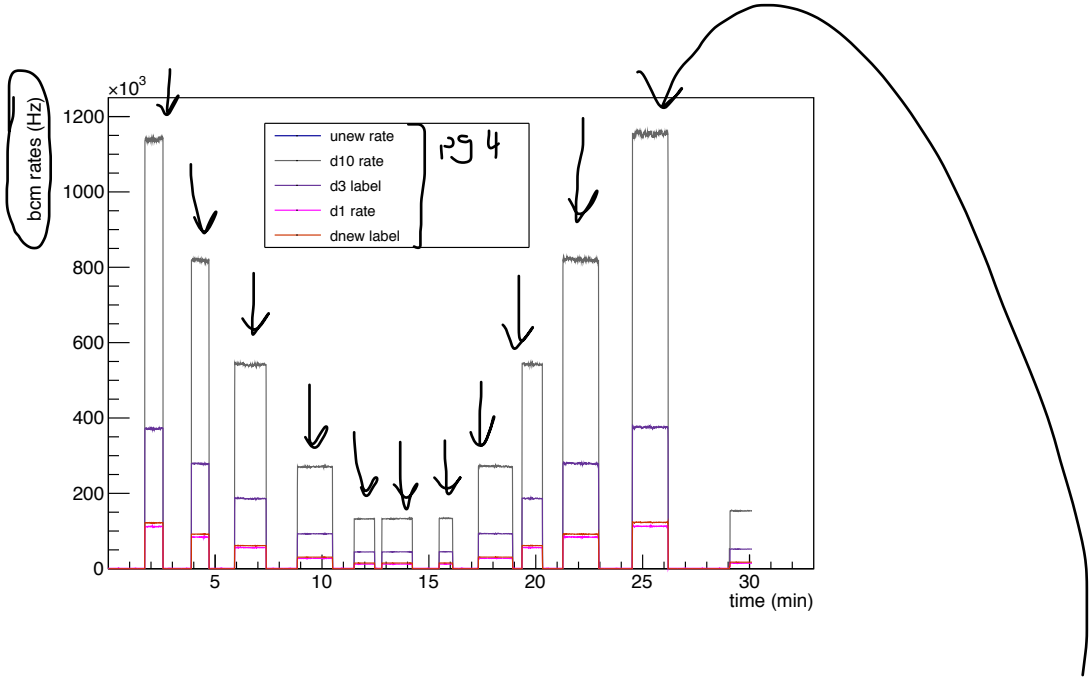


Figure 14: BCMs rates, Fall 2017.

Frequency average for each peak (current) is calculated

Unser - High Background, background subtraction needed

BCM - Low Background, background subtraction NOT needed

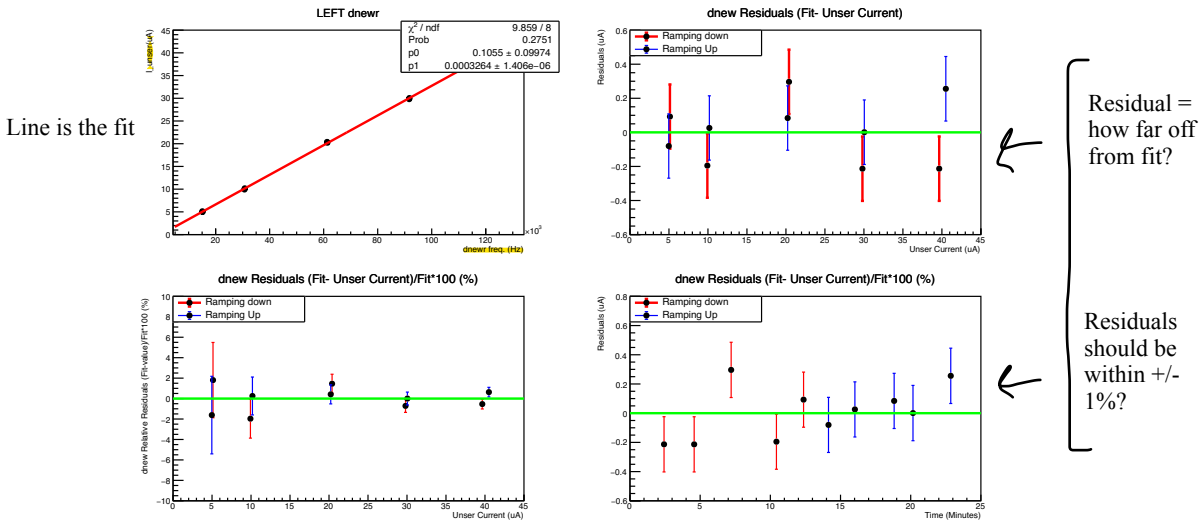
Unser current value is given by $\frac{1.5\mu A}{\sqrt{time}}$. How?

Since the BCMs are low noise devices, their calibration does not require background subtraction. Then, the frequency average for each current is calculated. Finally, the Unser current is plotted with respect to the average value of the BCMs frequency and a linear fit will give the parameters of the calibration.

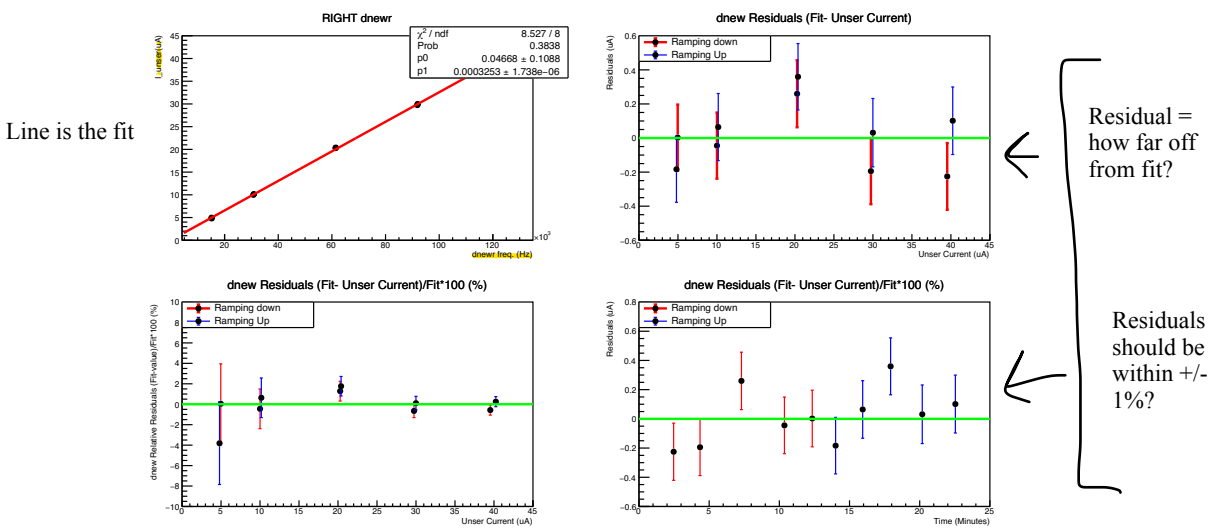
Where is this graph?

Figure 15 shows the dnew calibration for both arms. The parameters of the calibrations are given in the Table 2. It is important to note that during December 2017 the RHRS was not ramped up, therefore, the calibration was not needed since not data was taken with that spectrometer. Furthermore, the d10 signal was noisy for the RHRS during the first couple of weeks during the Fall of 2018, hence, the first calibration during that period was not available.

On the other hand, the settings values in the digital receivers for the unew signal changed during the winter break (As seen in the Table 2). While,the other signals remained within one percent difference. The digital settings are now recorded in the Archive.



(a) HRS dnew BCM calibration.



(b) RHRS dnew BCM calibration.

Figure 15: dnew BCM calibration.

	bcm signal	gain $\mu A/Hz$	gain error $\mu A/Hz$	offset μA	offset error μA	$corr_{11}$	$corr_{12}$	$corr_{21}$	corr
		2017-12-14 (Run 815) Calibration							
LHRS	d3	1.07E-04	4.63E-07	2.62E-01	9.97E-02	1	-8.13E-01	-8.13E-01	1
	d10	3.74E-05	3.85E-07	7.30E-03	1.38E-01	1	-8.08E-01	-8.08E-01	1
	dnew	3.26E-04	1.40E-06	1.05E-01	9.97E-02	1	-8.17E-01	-8.17E-01	1
	unew	3.95E-04	1.73E-06	1.38E-01	1.01E-01	1	-8.15E-01	-8.15E-01	1
		2018-01-25 (Run 1555) Calibration							
LHRS	d3	1.07E-04	7.89E-07	1.02E-01	9.60E-02	1	-9.01E-01	-9.01E-01	1
	d10	3.74E-05	3.57E-07	-4.76E-02	1.12E-01	1	-9.02E-01	-9.02E-01	1
	dnew	3.36E-04	2.74E-06	-9.75E-02	9.94E-02	1	-8.30E-01	-8.30E-01	1
	unew	3.00E-04	2.43E-06	-1.08E-01	9.94E-02	1	-8.30E-01	-8.30E-01	1
		2018-01-25 (Run 90882) Calibration							
RHRS	d3	1.07E-04	8.10E-07	1.11E-01	9.69E-02	1	-9.19E-01	-9.19E-01	1
	d10	3.74E-05	3.77E-07	-3.74E-02	1.14E-01	1	-9.20E-01	-9.20E-01	1
	dnew	3.35E-04	2.96E-06	-9.15E-02	1.01E-01	1	-8.55E-01	-8.55E-01	1
	unew	2.96E-04	2.60E-06	-1.04E-01	1.01E-01	1	-8.55E-01	-8.55E-01	1
		2018-05-04 (Runs 3099, 3100 and 3101) Calibration							
LHRS	d3	1.06E-04	1.06E-04	3.37E-01	1.81E-01	1	-9.36E-01	-9.36E-01	1
	d10	3.64E-05	4.32E-07	2.50E-01	1.80E-01	1	-9.37E-01	-9.37E-01	1
	dnew	3.45E-04	2.85E-06	-1.62E-03	1.09E-01	1	-8.55E-01	-8.55E-01	1
	unew	3.02E-04	2.50E-06	-3.77E-02	1.09E-01	1	-8.57E-01	-8.57E-01	1
		2018-05-04 (Runs 93067 and 93068) Calibration							
RHRS	d3	1.06E-04	1.24E-06	3.70E-01	1.72E-01	1	-9.43E-01	-9.43E-01	1
	d10	3.65E-05	4.82E-07	3.51E-01	2.03E-01	1	-9.55E-01	-9.55E-01	1
	dnew	3.49E-04	2.51E-06	-6.56E-02	9.32E-02	1	-8.53E-01	-8.53E-01	1
	unew	3.05E-04	2.21E-06	-1.08E-01	9.39E-02	1	-8.54E-01	-8.54E-01	1
		2018-10-05 (Run 3705) Calibration							
LHRS	d3	1.08E-04	1.14E-06	1.69E-01	1.41E-01	1	-9.05E-01	-9.05E-01	1
	d10	3.72E-06	3.96E-07	5.96E-02	1.42E-01	1	-9.07E-01	-9.07E-01	1
	dnew	3.30E-04	2.83E-06	-8.41E-02	1.03E-01	1	-8.56E-01	-8.56E-01	1
	unew	4.18E-04	3.58E-06	-9.21E-02	1.03E-01	1	-8.56E-01	-8.56E-01	1
		2018-10-05 (Run 93730) Calibration							
RHRS	d3	1.08E-04	1.15E-06	1.06E-01	1.42E-01	1	-9.05E-01	-9.05E-01	1
	d10	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx	xxxx
	dnew	3.29E-04	2.85E-06	-6.09E-02	1.03E-01	1	-8.55E-01	-8.55E-01	1
	unew	4.17E-04	3.60E-06	-6.86E-02	1.03E-01	1	-8.55E-01	-8.55E-01	1
		2018-10-23 (Run 3954) Calibration							
LHRS	d3	1.07E-04	1.22E-06	-2.11E-02	1.62E-01	1	-9.22E-01	-9.22E-01	1
	d10	3.69E-05	4.23E-07	-1.14E-01	1.63E-01	1	-9.23E-01	-9.23E-01	1
	dnew	3.25E-04	2.71E-06	-1.59E-01	1.05E-01	1	-8.52E-01	-8.52E-01	1
	unew	4.13E-04	3.44E-06	-1.57E-01	1.05E-01	1	-8.52E-01	-8.52E-01	1
		2018-10-23 (Run 94016) Calibration							
RHRS	d3	1.07E-04	1.23E-06	-3.90E-02	1.63E-01	1	-9.32E-01	-9.32E-01	1
	d10	3.70E-05	6.07E-07	-3.29E-01	2.31E-01	1	-9.24E-01	-9.24E-01	1
	dnew	3.26E-04	2.71E-06	-1.81E-01	1.05E+01	1	-8.53E-01	-8.53E-01	1
	unew	4.14E-04	3.44E-06	-1.78E-01	1.05E-01	1	-8.52E-01	-8.52E-01	1

What is corr?

Different calibration runs used

Table 2: BCM calibration constants from December 2017 to September 2018.

Constants generated from the Calibration runs?

6 Charge calculation

The charge is the integral of the current during the time of the run. The current can be calculated using the Equation 2, where I is the current, g and o are the gain and the offset given by the BCM calibration, and f is the BCM frequency. However, in a normal run the current trips several times, and the trips may add false negative charge values that will lead to a wrong charge value, due to the not linearity of the BCM's at low currents.

$$I = g \cdot f + o \tag{2}$$

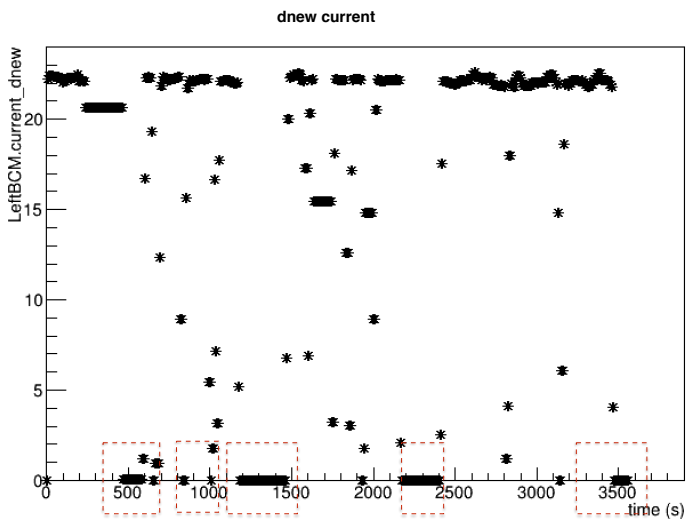


Figure 16: Current profile in an average run.

Additionally, the charge can be calculated using the Equation 3, where Q is the charge, g and o are the gain and the offset given by the BCM calibration, c are the BCM counts, and t is the time that the beam is on the target. In order to avoid the problems given by the beam trips, a safety current cut may be applied. In general, a very safe trip cut is required that the current is higher than 1 μA. In general, different currents may be applied, and that will be decided by the expert.

$$Q = g \cdot c + o \cdot t \tag{3}$$

Furthermore, an extra piece of information will be that in the Spring run the accelerator took took about 22 seconds to be back at 22 μA, after a beam trip.

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